



Genome analysis deciphered *Chryseobacterium indicum* is a distinct species associated with freshwater pufferfish

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ARTICLE INFO

Keywords:

Pufferfish

Chryseobacterium indicum sp. nov. PS-8^T

Genome analysis

Phylogenetic analysis

ABSTRACT

A bacterium, strain PS-8^T of the genus *Chryseobacterium*, was isolated from the skin of freshwater pufferfish (*Tetraodon cutcutia*). Strain PS-8^T is a Gram-negative, aerobic, non-motile, and rod-shaped bacterium. Colonies appear in yellowish-orange colors. The major cellular fatty acids were C15:0 iso, C17:0 iso 3OH, C15:0 iso 3OH, and C11:0 anteiso. The predominant polar lipids were phosphatidylethanolamine and amino lipids. The genome size is 4.83 Mb. The G + C content was 35.6%. The in silico dDDH homology, ANI, and AAI were below the cutoff value, 70% and 95% to 96%, respectively, suggesting that strain PS-8^T represents a defined species. The phylogenetic tree based on core and the non-recombinant genes showed the strain PS-8^T clustered with *Chryseobacterium gambrini* DSM 18014^T. Genome-wide analysis decodes several virulence factors of the genus *Chryseobacterium*, including genes for adherence, biofilm and stability, proliferation, resistance to immune response, and host-defense evasion system. The cladogram of the virulence genes showed a phylogenetic relationship among the *Chryseobacterium* species. Knowledge of the association of *Chryseobacterium* with freshwater pufferfish adds a new ecological niche to this bacterium.

1. Introduction

The members of the genus *Chryseobacterium* have been reported from various environments, including soil, phyllosphere, fish, industrial plants, and clinical samples (Behrendt et al., 2007; Li and Zhu, 2012; Zamora et al., 2012). Some species of *Chryseobacterium* isolated from patients causing diseases like emphysema, pneumonia, and meningitis (Bloch et al., 1997; Chiu et al., 2000; Izaguirre-Anariba and Sivapalan, 2020; Tsouvalas et al., 2020). In addition, *Chryseobacterium nematophagous* shows matrix-digesting properties such as collagenous matrices, feathers, and exoskeletons (Page et al., 2019). *Chryseobacterium* is not a critical pathogen for livestock animals and is not frequently associated with fish infections. However, there is a surge of clinical cases from diseased fish where *Chryseobacterium* sp. has been isolated. These include *Chryseobacterium scopthalmum*, *Chryseobacterium balustinum*, and *Chryseobacterium joostei* (Mudarris et al., 1994; Hugo et al., 2003). In the recent time, *Chryseobacterium piscicola* have been reported from external lesions of diseased farmed Atlantic salmon and rainbow trout in Chile (Ilardi et al., 2009), *Chryseobacterium viscerum*

was also found in gills and liver of the diseased rainbow trout (Zamora et al., 2012), and *Chryseobacterium arothri* from the kidneys of the pufferfish (*Arothron hispidus*) in Hawaii (Campbell et al., 2008). These reports suggest an emergence of novel *Chryseobacterium* spp. associated with aquaculture-reared fish worldwide (Loch and Faisal, 2015).

Chryseobacterium appeared as 1-2 mm diameter yellowish-orange colonies (Kirti et al., 2014; Venil et al., 2014). The genus *Chryseobacterium* is Gram-negative, rod-shaped and non-motile. G + C mol% of DNA range from 29 to 39% (Bernardet et al., 2005). The Major fatty acids are isoC15:0, isoC17:0 3OH, isoC15:0 3OH, and isoC15:0 2-OH, and polar lipids are phosphatidylethanolamine, amino lipids, and phospholipids (Kämpfer et al., 2009). During the investigation of bacteria associated with the healthy freshwater pufferfish (*Tetraodon cutcutia*) skin, strain PS-8 showing yellowish-orange colonies on nutrient agar (Difco) at 28 °C was isolated. We compared the genome of strain PS-8^T with 73 genome sequences of the type strains of the genus *Chryseobacterium* available in the NCBI database. The phylogeny based on the genome-wide core genes and evaluation of ANI and in silico dDDH suggested the strain PS-8^T is a distinct species. Additionally, we

Abbreviation: Average nucleotide identity, ANI; average amino acid identity, AAI; DNA–DNA hybridization, dDDH.

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<https://doi.org/10.1016/j.meegid.2024.105561>

Received 19 December 2023; Received in revised form 20 January 2024; Accepted 22 January 2024

Available online 26 January 2024

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predicted that the genes encode virulence factors in this group of bacteria.

2. Materials and methods

2.1. Isolation of bacterium and growth medium

Chryseobacterium strain PS-8^T was isolated from the mucus of the freshwater pufferfish skins, as described earlier (Das et al., 2022). The mucus from the pufferfish skins was collected with sterile cotton swabs and put into 1 mL sterile phosphate-buffered saline at pH 7.4. Mixed by vortexing and used as a master mix (Badhai et al., 2013). A 100 µl aliquot of the master mix was serially diluted using phosphate buffer saline (PBS), plated onto nutrient agar (BD, Difco), and incubated at 28 °C (Das et al., 2021). Several colonies that appeared on nutrient agar were picked and purified. A yellowish-orange colony, designated strain PS-8, was selected for further analyses. Cultures were maintained on nutrient agar (BD, Difco) and stored at 4 °C for short-term preservation. Cultures were stored at −80 °C in 15% (v/v) glycerol for long-term preservation.

2.2. Phenotypic characterization

Cell morphology and the presence of flagella were examined by transmission electron microscopy (JEOL, JEM-2100 Plus). Cells were grown on a nutrient agar plate for 24 h. With a sterile toothpick, a few colonies were transferred to the phosphate buffer saline (1 x PBS) to make a bacterial suspension (~10⁹ 1/mL). One drop of cell suspension was applied to a carbon-coated grid (300 mesh, 3 mm diameter) for one min., washed with sterile distilled water (3 × 30 s), and finally negatively stained with 2% phosphotungstic acid for 30 s. Excess fluid was removed using filter paper, and the grids were examined under a microscope (Jyoti et al., 2010). Gram staining was done using a commercial kit (Becton–Dickinson). Oxidase activity was assayed with a commercial kit (Hi-Media). Hydrogen peroxide (10%, v/v) was used for the catalase activity. Anaerobic growth was tested using the BD GasPak EZ system (Becton Dickinson). Growth physiology at different pH ranges (4.0–10.0; in steps of 0.5 pH units) was examined in nutrient broth prepared in buffered solutions (pH 5.0–5.5, 50 mmol/L acetic acid–50 mmol/L sodium acetate; pH 6.0–8.0, 50 mmol/L KH₂PO₄–50 mmol/L K₂HPO₄; pH 8.5–10.0, 50 mmol/L Na₂CO₃–50 mmol/L NaHCO₃) at 30 °C for 3 days (Jung et al., 2012). Growth at different temperatures was monitored by measuring OD₆₀₀ with a spectrophotometer (CARY 300, UV–Vis spectrophotometer, Varian). The growth in different concentrations of NaCl (0–15%) was determined in MM63 minimal media and incubated at 28 °C for two days (Jiang et al., 2014). Phenotypic characteristics such as Voges Proskauer test, indole, methyl red, H₂S production, citrate utilization, nitrate reduction, phenylalanine deaminase, hydrolysis of esculin, lysine decarboxylase, ornithine decarboxylase, arginine dihydrolase, urease, casein, DNA, gelatin, starch and Tween 20 and 80 were tested according to Mata et al. (2002). Phenotypic characterization was also performed using API 20 NE, API 20E, API ZYM (bioMe'rieux), and Biolog GNII microplates. The hemolytic activity was tested on Columbia blood agar (Das et al., 2022). Susceptibility to different antimicrobials was determined by the disc diffusion method on nutrient agar medium (Kumari et al., 2014). The size of the antimicrobial discs was 4.0 mm contained the following antimicrobial discs (µg/disc): kanamycin (30 µg), tetracycline (30 µg), rifampicin (5 µg), spectinomycin (100 µg), gentamycin (10 µg), ampicillin (10 µg), erythromycin (15 µg), chloramphenicol (30 µg), nalidixic acid (30 µg), ciprofloxacin (5 µg), streptomycin (10 µg), polymyxin B (300 µg), vancomycin (30 µg) and trimethoprim (5 µg). Sensitivity to antimicrobials was recorded after 48 h of incubation by measuring the mean diameter of the inhibition zone. Further, the differential growth responses of the strain PS-8^T was compared with closely related strain *C. gambrini* DSM 18014^T (16S rRNA similarity 99.17%). The phenotypic analyses of the

Table 1

Phenotypic characteristics of *Chryseobacterium indicum* PS-8^T and closely related strain *C. gambrini* DSM 18014^T. Data were obtained in this study unless indicated. Data from * Herzog et al. (2008).

Characteristics	C. strain PS-8 ^T	*C. gambrini DSM 18014 ^T
API 20E		
Tryptophan deaminase	+	–
Fermentation of Glucose	+	–
Fermentation of Mannose	–	+
Fermentation of Inositol	–	+
Fermentation of Sucrose	+	–
Fermentation of Arabinose	+	–
API 20 NE		
L-Arginine	–	+
D-Glucose	–	+
L-Arabinose	–	+
D-Mannose	–	+
D-Maltose	–	+
Adipic acid	–	+
Phenyl acetic acid	–	+
BIOLOG GNII		
α-Cyclodextrin	+	–
D-Mannitol	–	+
D-Mannose	–	+
D-Melibiose	+	–
β-Methyl-D-Glucoside	+	–
D-Psicose	+	–
D-Sorbitol	+	–
Methyl Pyruvate	–	+
D-Galacturonic Acid	–	+
D-Glucuronic Acid	–	+
α-Keto Butyric acid	+	–
α-Keto Valeric acid	+	–
D,L-Lactic acid	–	+
Malonic acid	+	–
Propionic Acid	+	–
Glucuronamide	–	+
L-Pyrogutamic Acid	+	–
D-Serine	–	+
Inosine	+	–
Ampicillin (10 µg)	–	+
Kanamycin (30 µg)	+	–

Positive (+), negative (–).

strain PS-8^T are shown in Table 1.

2.3. Chemotaxonomic analysis

Cells were grown in nutrient broth (Difco) to mid-exponential phase on a rotary shaker at 28 °C for the lipid analysis. Polar lipids were extracted from 200 mg dry cells described by Bligh and Dyer (1959) and separated by two-dimensional TLC (silica gel 60 F254, cat no. 1.05554.0007; Merck). The solvent system used was described earlier (Kumari et al., 2014). Diaminopimelic acid (DAP) and cell wall sugar were detected following Stanek and Roberts (1974) methods. Cellular fatty acids were analyzed from cells grown on nutrient agar (Difco) plates at 30 °C for three days. Cells were saponified, and the fatty acid methyl ester mixtures were separated using the Sherlock Microbial identification system (MIDI, Microbial ID), consisting of an Agilent model 6890 N gas chromatograph.

2.4. Whole genome sequencing and annotation

The genomic DNA was isolated using the methods of Sambrook and Russell (2001). The DNA concentration and quality were measured using a NanoDrop 8000 spectrophotometer (Thermo Fisher Scientific). A combination of short-read Illumina and long-read Oxford Nanopore Technologies sequencing platforms was used to generate the complete genome sequence of *Chryseobacterium indicum* strain PS-8^T. Illumina

short-read DNA sequencing was carried out using the Illumina HiSeq 4000 next-generation sequencing platform to produce paired-end sequence reads with 150-bp read length. Read quality was assessed with the FastQC v0.11.5 program (<http://www.bioinformatics.babraham.ac.uk/projects/fastqc>) using default parameters. Low-quality bases and adapters were removed with Trimmomatic v0.36 (Bolger et al., 2014) using the parameters mentioned in the manual. Trimmed reads were assembled de novo using Unicycler v0.4.7 (Wick et al., 2017), followed by assembly polishing with Pilon v1.23 (Walker et al., 2014). Assembly quality was assessed with QUAST v4.6.1 (Gurevich et al., 2013). A genomic library was prepared using the Nanopore ligation sequencing kit (SQK-LSK109; Oxford Nanopore Technologies, Oxford, UK) for long-read Nanopore sequencing. The library was sequenced with an R9.4.1 MinION flow cell (FLO-MIN106) using MinKNOW v2.0 with default settings. Barcode and adapter sequences were trimmed from Nanopore long reads using Porechop v0.2 (<https://github.com/rrwick/Porechop>), and reads with a minimum length of 1 kb were filtered using seqtk v1.2 (<https://github.com/lh3/seqtk>) for downstream analysis. The hybrid genome assembly was performed using Unicycler v0.4.9 in hybrid assembly mode (Wick et al., 2017). The highly accurate Illumina short reads were aligned against the long Nanopore reads to sort out random sequencing errors (Wick et al., 2017). The assembled genomes were annotated using the NCBI Prokaryotic Genome Annotation Pipeline (PGAP) v4.9 with default parameters (Tatusova et al., 2016). The completeness and contamination of the whole-genome sequence were measured using CheckM (Parks et al., 2015). The genomic G + C content and assembly statistics were determined using a Perl script (https://github.com/tomdeman-bio/Sequence-scripts/blob/master/calc_N50_GC_genomesize.pl). Additionally, the RAST Server operates the GLIMMER algorithm which was used to predict the protein-coding genes (Salzberg et al., 1998; Overbeek et al., 2014).

2.5. Comparative genomics and phylogenetic analysis

The genomic relatedness of strain PS-8^T was compared with the reference genomes of *Chryseobacterium* available in the National Centre for Biotechnology Information (NCBI) database. The average nucleotide identity (ANI) was calculated using the Python module pyani (<https://github.com/widdowquinn/pyani>) with the ANiB method. In silico dDDH similarity was measured with the help of the Genome-to-Genome distance calculator (formula 2) (Meier-Kolthoff et al., 2013). Average amino acid identity (AAI) was estimated using the 'aai_wf' function implemented in the compareM program (<https://github.com/dparks1134/CompareM>). For phylogenomic analysis using the genome-wide core genes, the type strains with validly named species of *Chryseobacterium* with >95% genome completeness were retrieved from the NCBI database (<https://github.com/kblin/ncbi-genome-download/>). The core genes extracted by the UBCG pipeline using default parameters (Na et al., 2018) were concatenated, and a maximum-likelihood tree was constructed with the GTR model using the RAxML tool (Stamatakis, 2014). Furthermore, a non-recombinant core-genome based phylogenetic tree was reconstructed following the method of Mateo-Estrada et al. (2019).

2.6. Prediction of virulence genes

Blast and reciprocal blast (blastp) (Altschul et al., 1990) were performed (query coverage, percentage identity $\geq 40\%$) with clinically verified core data-set (3580 genes) of virulent factor database (VFDB) (Chen et al., 2005) to retrieve the putative virulence genes of *Chryseobacterium indicum* strain PS-8^T and the type strains compared in this study. The cladogram-heatmap was generated using the ETE-3 package of the Python program. Hits with either coverage or identity lower than 40% were not considered.

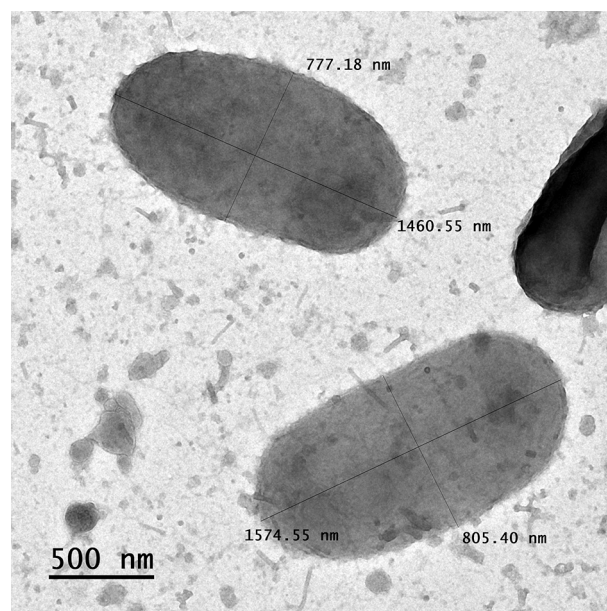


Fig. 1. Electron micrograph of a cell of *Chryseobacterium indicum* PS-8^T.

3. Results and discussion

3.1. Phenotypic characterization

The strain PS-8^T appeared yellowish-orange, with smooth colonies up to 1.0 mm in diameter. The strain was Gram-reaction-negative, strictly aerobic, and non-motile, with rod-shaped cells approximately 0.7–0.8 μm wide and 1.4–1.5 μm long in size (Fig. 1). Growth temperature in nutrient broth ranges from 10 to 40 $^{\circ}\text{C}$, with optimal growth at 28–37 $^{\circ}\text{C}$. The pH range for growth is between 4.5 and 11, and the optimum growth is at a pH of 5.5–7.5. Strain PS-8^T could grow in the presence of 0–6% NaCl (w/v), and the optimal growth occurred at 0.5% NaCl. The strain PS-8^T was negative for hemolytic activity. Strain PS-8^T is susceptible to vancomycin (30 μg), tetracycline (30 μg), rifampicin (5 μg), spectinomycin (100 μg), gentamycin (10 μg), ampicillin (10 μg), trimethoprim (5 μg), erythromycin (15 μg), chloramphenicol (30 μg), nalidixic acid (30 μg), ciprofloxacin (5 μg), streptomycin (10 μg) and resistant to kanamycin (30 μg), polymyxin B (300 μg). The different growth responses of the strain PS-8^T are shown in (Table 1).

3.2. Chemotaxonomic characterization

The polar lipid spectrum contained different lipids, including phosphatidylethanolamine (PE), four amino lipids (AL1–4), and nine unknown lipids (L1–9) (Fig. S1). The lipid profile of strain PS-8^T was similar to that of the reference strains of the genus *Chryseobacterium* (Montero-Calasanza et al., 2014). The major cellular fatty acids of strain PS-8^T were 15:0 iso (38.04%), 17:0 iso 3OH (16.51%), 15:0 iso 3OH (3.15%), and 11:0 anteiso (2.38%) (Table S1). The fatty acid composition of strain PS-8^T is consistent with the previous results for recognized species of *Chryseobacterium*, justifying its placement in this genus (Kämpfer et al., 2009). Strain PS-8^T differed from the reference strain by excluding 15:0 iso 2OH, 16:0 3OH, 17:1 iso cis9, and 17:0 cyclo (Table S1). The cell wall sugars contained glucose, galactose, and ribose. The diamino acid in peptidoglycan is meso-diaminopimelic. The overall chemotaxonomic characteristics of strain PS-8^T support its placement in the genus *Chryseobacterium*.

3.3. Genome features

The genome of the strain PS-8^T is 4.83 Mb. The genomic G + C

Table 2
Comparison of the genomic characteristics of *Chryseobacterium indicum* PS-8^T with closely related species of *Chryseobacterium*.

Sl. No.	Strain	Accession	Size (MB)	G + C (%)	16S rRNA similarity (%)	ANIb (%)	AAI (%)	in silico DDH (%)
1.	<i>C. strain</i> PS-8 ^T	JACSGT000000000	4.83	35.6	100	100	100	100
2.	<i>C. gambrini</i> DSM 18014 ^T	FTOV000000000	4.84	36.1	99.17	92.19	92.73	47.9

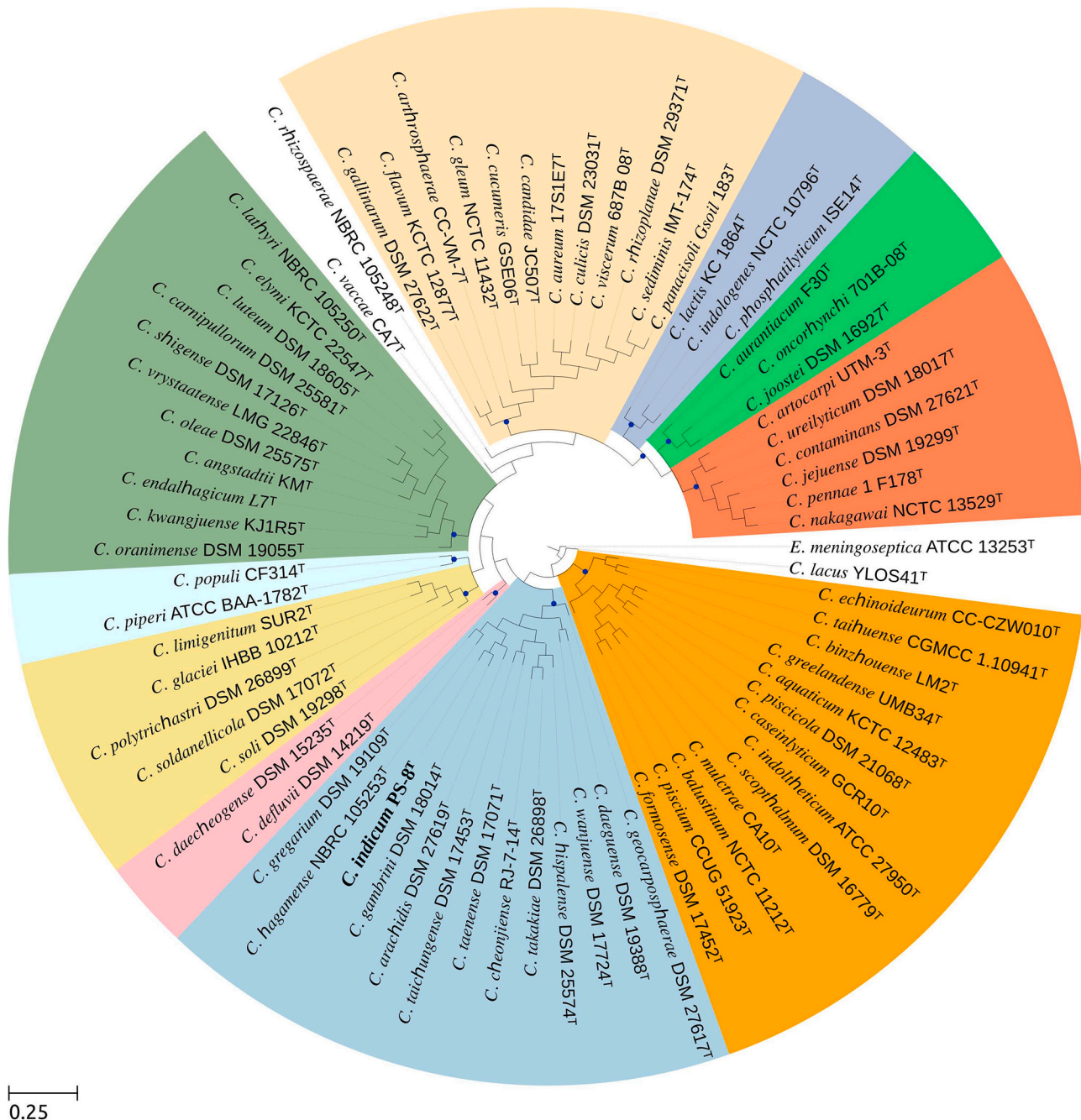


Fig. 2. Maximum-likelihood phylogenetic tree based on core gene sequences of *Chryseobacterium indicum* PS-8^T with available genome sequences of type species. Bootstrap values >62% for all the branches based on 1000 resampling. Bar, 0.25 substitutions per nucleotide position.

content was 35.6%. The genomic G + C content value is within the range of the *Chryseobacterium* species. The de novo assembly produced five scaffolds covering 4,837,804 bp (85.983%). The genome contained 90tRNA, five 5SrRNA, five 16S rRNA, and five 23S rRNA. Further, we

determined the genome-wide genes connected to subsystems and their distribution in different functional categories for the strain PS-8^T and *Chryseobacterium gambrini* DSM 18014^T. As a result, 4595 protein-coding genes from strain PS-8^T and 4627 genes from *Chryseobacterium gambrini*

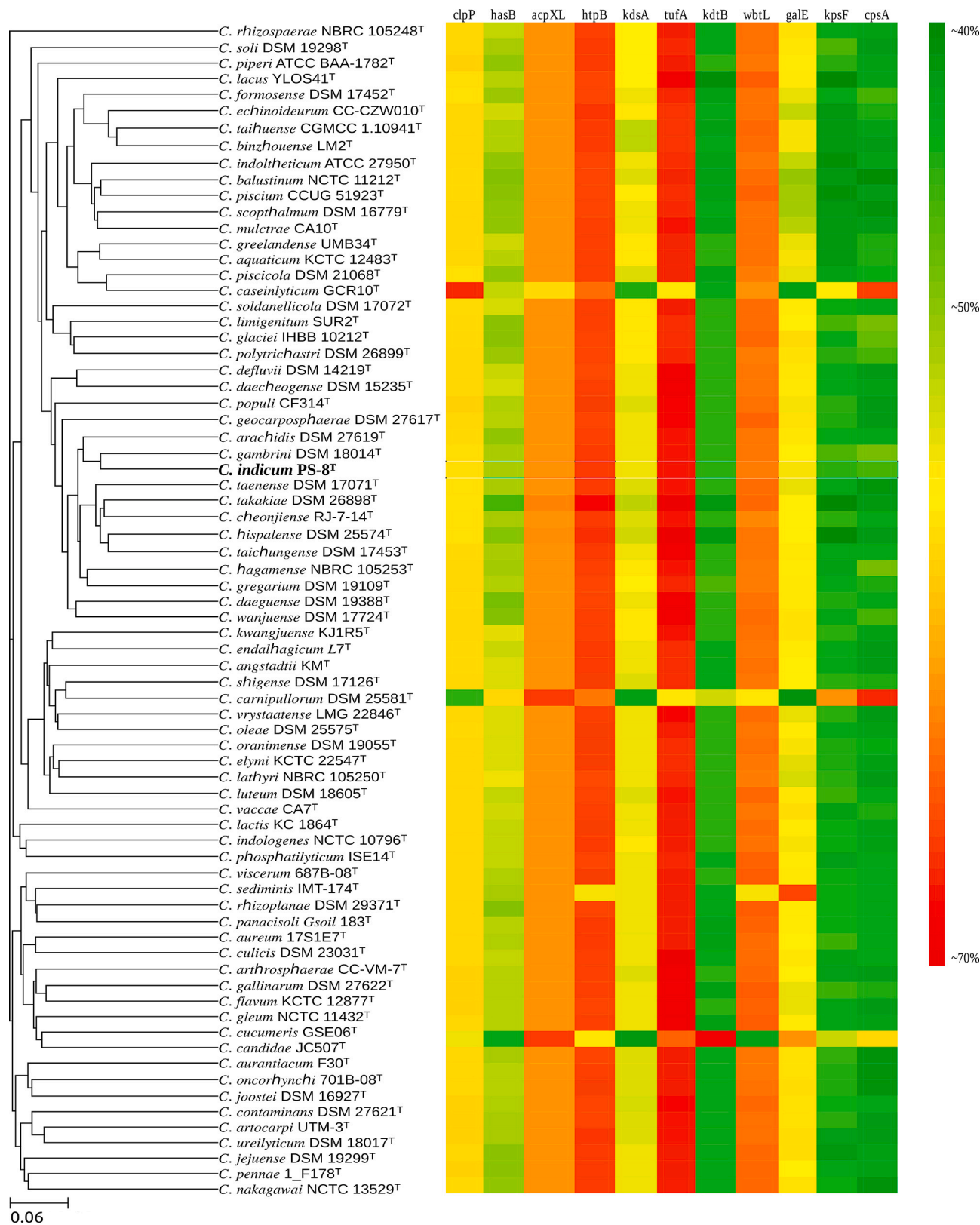


Fig. 3. Cladogram-heatmap based upon protein sequences of virulence genes in *Chryseobacterium*. Conservation in nucleotide sequences of orthologous virulence genes in *Chryseobacterium indicum* PS-8^T with reference strains. Bar represents percent nucleotide sequence identity.

DSM 18014^T were predicted. Of these 924 of strain PS-8^T and 921 genes of *Chryseobacterium gambrini* DSM 18014^T could be annotated by RAST's automated homology analysis procedure and assigned to functional categories (Fig. S2). The unique gene features of *Chryseobacterium*

strain PS-8^T compared with strain *Chryseobacterium gambrini* DSM 18014^T are aspartate racemase (EC 5.1.1.13), threonine dehydrogenase and related Zn-dependent dehydrogenases, endo-1,4-beta-xylanase A precursor (EC 3.2.1.8), alkylated DNA repair protein AlkB, type III

restriction-modification system DNA endonuclease res (EC 3.1.21.5), type III restriction-modification system methylation subunit (EC 2.1.1.72), lanthionine biosynthesis protein LanB, heavy metal RND efflux outer membrane protein, CzcC family, etc. (Table S2).

3.4. Overall genome-related indices and phylogenetic analysis

Whole-genome sequencing is the current gold standard for the taxonomic and phylogenetic investigation of microorganisms. Therefore, we evaluated the overall genome relatedness index (OGRI) with the type strains that showed 16S rRNA gene sequence similarity of >98% to delineate *Chryseobacterium indicum* strain PS-8^T as a defined species of the genus *Chryseobacterium*. The average nucleotide identity (ANI) and average amino acid identity (AAI) values between PS-8^T and the validly named reference species (Table S3) were below the threshold value (95–96%), justifying bacterial species delineation (Konstantinidis and Tiedje, 2005a; Konstantinidis and Tiedje, 2005b). Thus, strain PS-8^T represents a new species in the genus *Chryseobacterium*. Further, in silico genomic dDDH similarity values were lower than the 70% cut-off to define bacterial species (Meier-Kolthoff et al., 2013). Therefore, the ANI, AAI, and dDDH data indicate that the strain PS-8^T can be proposed as a defined species in the genus *Chryseobacterium* (Table 2). Furthermore, the essential component of the current taxonomy is the genome-wide comparative phylogenomic analysis based on core genes superior to those found on single-gene markers. Analysis based on a single gene marker such as 16S rRNA gene sequence suffers from well-known drawbacks such as a low resolution or bias due to homologous recombination. The 16S rRNA gene sequences shared 99.17% similarity with the closely related type strain *Chryseobacterium gambrini* DSM 18014^T, which is beyond the 98.7% threshold. Therefore, phylogenomic analysis of the *Chryseobacterium indicum* PS-8^T strain was performed using core genes of the type strains in the NCBI database. The maximum-likelihood (ML) phylogenetic tree based on core genes showed the strain PS-8^T clustered with *Chryseobacterium gambrini* DSM 18014^T. It formed a distinct clade (Fig. 2), which is similar to the phylogenetic tree based on the non-recombinant genes (Fig. S4), indicating the robustness of the tree topology and justifying *Chryseobacterium indicum* PS-8^T sp. nov.

3.5. Analysis of virulence genes

A total of 41 virulence-related genes were predicted in the genome of the *Chryseobacterium indicum* PS-8^T (Fig. S5). The virulence genes identified showed closeness to genes that reveal functions like (i) adherence, biofilm, and stability (*glf*, *msrA/B*, *pilB*, *acpXL*, *lpxD*, *clpE*, *cpsO*, *clpE*, *lfpA*); (ii) proliferation (*tufA*, *cap8E*, *icl*, *manC*, *fleQ/ftrc*, *galE*, *kfc*, *bioB*, *clpP*, *kdtB*, *csgD*, *carB*); (iii) resistance to immune response (*htpB*, *katA*, *wbtL*, *cj1136*, *hasB*, *kdsA*, *lirB*, *gnd*, *clbC*, *cylG*, *sodB*, *kpsF*) and (iv) host-defense evasion (*kpst*, *clpE*, *kfc*, *ybtQ*, *cap8G*, *clpP*, *FTT-0798*) (Fig. S6). Considering identity cut-off (40%), 11 virulence genes (*clpP*, *hasB*, *acpXL*, *htpB*, *kdsA*, *tufA*, *katB*, *wbtL*, *galE*, *kpsE*, *cpsA*) from the strain PS-8 were selected for comparison with the 73 type species of *Chryseobacterium*. The cladogram and the heatmap based on the virulence genes showed the closeness among the *Chryseobacterium* species (Fig. 3).

3.6. Description of *Chryseobacterium indicum* sp. nov.

Chryseobacterium indicum sp. nov. (in 'di.cum. L. neut. Adj. *indicum*, of or belonging to India, where the type strain was isolated).

Strain PS-8^T is Gram-negative, aerobic, non-motile, rod-shaped, approximately 0.7–0.8 µm wide and 1.4–1.5 µm long. Colonies of strain PS-8^T appeared a yellowish-orange color. Growth occurred at 10–40 °C and pH 5–10. Optimal growth was observed at 28–30 °C and pH 6.5–7. It can grow at an optimum concentration of 0.5% NaCl (w/v). Positive for catalase, oxidase, hydrolysis of gelatin, DNA, casein, aesculin, tween 40 and tween 80 and negative for arginine dihydrolase, lysine

decarboxylase, production of H₂S, the activity of galactopyranosidase, ornithine decarboxylase, tryptophan deaminase, Voges-Proskauer, urease, citrate utilization, hydrolysis of starch, and cellulose, production of acetoin, nitrate, and nitrite reduction. Acid is produced from sucrose, arabinose, and glucose. Predominant fatty acids are 15:0 iso, 17:0 iso 3OH, 15:0 iso 3OH, and 11:0 anteiso. Polar lipids include phosphatidylethanolamine and amino lipids. The G + C content is 35.6%.

The type strain is PS-8^T (= TBRC 15233^T = NBRC 115235^T), which was isolated from the skin of freshwater pufferfish.

4. Conclusion

The phylogenomic and chemotaxonomic study revealed that strain PS-8^T could be a defined species. Additionally, genome analysis predicted genes encoding several virulence factors, which may help in understanding the pathogenicity of this group of bacteria.

Funding

S.K.D was supported in part by the funding received from Department of Biotechnology, Government of India (D.O.No. BT/PR7661/AAQ/3/629/2013).

Ethics approval

This study was carried out with approval from the Institutional Animal Ethics Committee (Letter No. V-311-MISC/2017–18/ILS/884).

CRediT authorship contribution statement

Manish Prakash Victor: Methodology, Formal analysis, Data curation. **Ritu Rani Archana Kujur:** Writing – review & editing, Validation, Methodology, Formal analysis. **Lipika Das:** Methodology, Investigation, Formal analysis. **Subrata K. Das:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare no competing interests.

Data availability

All data generated or analyzed during this study are included in this article. The GenBank/EMBL/DBJ accession number for the genome and 16S rRNA gene sequences of *Chryseobacterium* strain PS-8^T are JACSGT000000000 and MZ305332, respectively.

Acknowledgments

The authors M.P.V, LD and RRAK acknowledge the *India-UK* research grant funded by Newton-Bhabha Fund and Council of Scientific and Industrial Research (CSIR), New Delhi, Government of India, respectively for providing the research fellowship. We acknowledge the Distributed Information Sub-Center (DISC) at the Institute of Life Sciences, Bhubaneswar, for the computational facility and Transmission Electron Microscopic Facility. We thank Professor Bernhard Schink (Universitaet Konstanz, Germany) for his etymological advice.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.meegid.2024.105561>.

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